COMP3012

Digital Forensics and Malware Analysis

Report

10745379

Contents

Word Count 1
1. Introduction 2
2. Acquisition 2
2.1 Methodology 2
2.2 Research Techniques 3
2.3 Analysis Methods 3
2.4 Considerations 4
3. Examination & Analysis 5
3.1 Analysis Tools 5
3.1.1 Autopsy 5
3.1.2 FTK 5
3.1.3 Volatility 5
3.1.4 CAINE 6
3.1.5 ProDiscover 6
3.2 Analysis 6
3.2.1 Examination & Analysis Using Forensic Tools 6
3.2.2 Justification 8
3.2.3 Operational Trade-Offs 8
3.2.4 Findings 8
4. Conclusion 8
5. References 9

Word Count

2828

1. Introduction

This report presents a detailed tender for a digital forensic investigation tailored to the scenario presented with Acme Bank, by integrating tools like Autopsy, Forensic Toolkit (FTK), Scalpel and Sleuth Kit. The investigation follows key phases: Identification, Preservation, Collection, Examination, Analysis and Presentation. Identification involves documenting the evidence such as the compromised PC and keylogger, while Preservation ensures evidence integrity through isolation and forensic imaging facilitated by the forensic tools mentioned. Collection consists of gathering data from various sources, such as the network and firewall logs. The examination phase involves analysing the evidence, including keystroke logs, for indicators of compromise, utilising Autopsy for efficient data parsing and FTK for comprehensive analysis. Analysis aims to correlate findings to reconstruct the incident sequence. Operational trade-offs, like balancing thoroughness with the Acme's business continuity, are considered, and the likelihood of finding relevant data depends on factors like attacker sophistication and incident duration. This structured approach, leveraging relevant forensic tools ensures a detailed investigation to aiding Acme in understanding the incident and taking appropriate actions.

Including Autopsy, FTK, Sleuth Kit and Scalpel in digital forensic development offers numerous benefits. Sleuth Kit delivers a robust suite of command-line tools for forensic analysis, allowing tasks like file system analysis, timeline generation, and keyword searching. Its open-source nature ensure adaptability to diverse investigation needs. Autopsy, a graphical interface built upon Sleuth Kit, enhances the investigation process with its user-friendly environment. It supports forensic image analysis, keyword searches, and data visualisation while automating repetitive tasks for increased efficiency. Additionally, they support various file systems and disk formats, ensuring compatibility with various digital environments encountered during investigation. FTK further enriches this with its comprehensive forensic analysis capabilities, offering advanced deep data examination and analysis features. Integrating these tools empowers Acme to conduct meticulous analysis of the digital evidence, contributing to the investigation process's identification, preservation, and analysis phases.

2. Acquisition

2.1 Methodology

The analysis methodology implemented in this investigation encompasses a series of critical steps designed to examine digital evidence comprehensively. Initially, the collected data, which includes system logs, network traffic records and forensic images, undergoes meticulous inspection using advanced tools such as Autopsy, Sleuth Kit, FTK and Scalpel. They are instrumental in identifying potential compromise indicators (Yaacoub, 2021). Keyword searching and timeline analysis techniques are then employed to reveal relevant information embedded within the data. Additionally, data carving techniques are utilised to recover deleted or hidden files, thus expanding the scope of the investigation. Furthermore, correlation and pattern recognition play pivotal roles in constructing the sequence of events

leading to the incident, aiding in identifying malicious activity. Throughout the analysis process, most attention is devoted to maintaining the integrity of evidence and strictly adhering to forensic best practices to ensure credibility of findings in legal proceedings. Finally, investigation findings are meticulously documented and presented clearly and concisely, ensuring stakeholders fully grasp the implications of the analysis results. This approach facilitates informed decision-making and enhances the effectiveness of incident response efforts.

Data collection from various sources typically refers to gathering data from different sources within the company's digital environment. This can include data from systems, servers, other network devices, logs, and any other sources that could provide insights into the incident being investigated. Ensuring to acquire the compromised staff PC and keylogger early on. Another aspect to consider may be to perform on-site investigation after closing hours or at night to avoid raising suspicion with employees and customers.

2.2 Research Techniques

Preparation for the investigation involves thorough understanding of the incident's scope and objectives, with necessary permissions obtained for system and data access. Forensic tools required are configured and tested to meet investigation requirements. Through preparation anticipates and mitigates potential obstacles, ensuring a smooth and successful outcome (Akbal, 2021). Personnel undergo training on evidence handling and chain of custody protocols to maintain data integrity. Communication channels and reporting mechanisms are established for regular updates and collaboration.

The extraction and identification phases play pivotal roles in digital forensic investigations, emphasising collecting and recognising fitting evidence. In the extraction phase, data is retrieved from the various sources, utilising forensic tools. The process includes creating forensic images of packing media to preserve indication integrity and prevent alteration. Concurrently, network traffic logs and system event logs are collected to capture potential indicators of compromise. In the identification phase, the extracted data undergoes examination to pinpoint evidence relevant to the investigation's objectives. It consists of identifying files containing sensitive information, such as login credentials or suspicious activity logs, through keyword searches, file signature analysis, and metadata examination. Data hashing techniques may also be employed to verify integrity of extracted files. By systematically extracting and identifying relevant evidence, forensic investigators establish the foundation for subsequent analysis and examination phases, facilitating the discovery of critical insights into the incident.

2.3 Analysis Methods

Analysis in digital forensic investigations involve employing various techniques such as timeline analysis, correlation of events, and pattern recognition. Autopsy facilities this process through graphical representations of file systems, timelines, and search results, enhancing the efficiency of detecting anomalies and suspicious activities. Additionally, Scalpel, a file carving tool, excels in extracting data from fragmented or damages files, providing forensic investigators with valuable evidence. Moreover, other tools with unique features are

beneficial, e.g. Volatility has memory analysis capabilities that can uncover attacker activities not observable through traditional file-based analysis. Maltego, an OSINT tool, helps visualise relationships between entities, such as IP addresses and social media profiles, aiding in the investigation. By integrating these tools and techniques, it is possible to reconstruct the sequence of events, discern the attacker's strategy and evaluate the impact on the organisation's infrastructure and data integrity, thus delivering actionable insights.

Timeline Analysis involves arranging chronological occurrences from system logs and network traffic to reconstruct events preceding the incident, aiding in identifying patterns and areas of compromise (Javed, 2022). The correlation of events connects pieces of evidence to uncover malicious activity; aligning login timestamps with firewall logs reveals unauthorised access. Pattern Recognition identifies recurring patterns indicative of cyber threats, aiding in determining the nature and source of attacks (Bhuiyan, 2024). Keyword Searching pinpoints relevant evidence within large datasets by identifying specific terms or phrases related to known attack techniques or malware, aiding analysis (Ozcan, 2021). These methods collectively enhance the investigation by providing insights into incident timelines, behaviour patterns and evidence identification. Further discussed in Section 3.

2.4 Considerations

Considerations include evidence integrity, contextual understanding, collaboration, estimate total cost and timeline. Utilising tools like Autopsy, FTK and Scalpel, though costly, ensures balance between thoroughness and efficiency, supporting the investigation's objectives effectively.

Evidence Integrity: Throughout acquisition and analysis, maintaining evidence integrity remains paramount. Storing evidence strictly adhering to forensic standards, including meticulously documenting the chain of custody and preserving original evidence is imperative. These practices are crucial to ensure credibility and admissibility of findings in legal proceedings (Moric, 2021).

Contextual Understanding: Accurate interpretation of finding requires investigators to understand Acme's systems, network architecture, and security policies thoroughly. It's essential to consider the unique circumstances and constraints of the environment being investigated to contextualise analysis efficiently.

Collaboration and Communication: Effective collaboration and communication among the investigation team members and stakeholders is essential for sharing findings, discussing hypotheses, and making informed decisions. Consistent updates and status reports promote transparency and accountability, fostering a productive, investigation process.

Cost Estimation: The investigation's cost varies based on the scope and resources. The estimate should include consultation, tool acquisition, personnel, training, miscellaneous expenses and duration of investigation, totalling £x - £x.

Miscellaneous costs may include travel expenses for on-site investigations, accommodation for personnel, and incidentals such as printing and documentation. Additionally, personnel

costs may cover forensic investigators, analysts, legal experts, and support staff involved in the investigation, accounting for their time and expertise.

* *Cost estimates provide a guideline and may be adjusted based on investigation needs and conditions faced.*

3. Examination & Analysis

3.1 Analysis Tools

3.1.1 Autopsy

Integrating Autopsy in digital forensic investigations marks a significant advancement, particularly in the detection and correlation of Brute Force attacks. By utilising Autopsy's robust keyword search functionality and tagging features, couples with manual examination of log data, forensic specialists can systematically identify and analyse malicious login attempts and their associated IP addresses (Almutairi, 2020). This integrated approach streamlines the investigative process, enabling to uncover relevant evidence efficiently. Moreover, Autopsy's competence to produce Forensic Lucid reports enhance the clarity and comprehensiveness of findings, strengthening the efficacy of cyber security incident management and supporting legal proceedings. With its versatile tools and user-friendly interface, massive datasets can be effectively managed, ultimately allowing strengthening the establishment's capability to proactively detect and respond to security threats.

3.1.2 FTK

The Forensic Toolkit (FTK) is a robust electronic inquiry tool widely used by data security, technology, and law implementation experts. Custom-made for post-mortem computer forensics, FTK specialises in acquiring and analysing diverse digital devices, ranging from hard drives to flash memory and mobile devices (Al Hosani, 2020). FTK has the ability to create precise forensic images, ensuring data integrity throughout the investigative process. Moreover, it conducts comprehensive analysis including email parsing, file decryption, data carving, and visualisation, enabling to extract valuable insights from digital evidence swiftly and accurately. Some distinct features include Cerbus for automated malware detection and Optical Character Recognition (OCR) capabilities for converting images into readable text, further augmenting FTK's versatility and advanced functionalities. It is an invaluable tool, facilitating efficient data analysis and evidence presentation to effectively support this investigation.

3.1.3 Volatility

The Volatility Framework is a pivotal tool primarily employed for memory analysis umps. Offering an open-source collection of Python tools under the GNU General Public Liscence, it provides a versatile solution with distinct profiles tailored for various operating systems, including Windows, Mac, and Linux. Through commands like 'image info', which verifies the OS profile, and specific analyses such as 'amcache' for Am Cache info containing program execution metadata or 'pslist' for process enumeration, Volatility reveals potential, malicious

activities embedded within memory dumps. Moreover, Volatility Workbench, an integrated GUI framework, enhances user-friendliness and speed in conducting forensic analyses. Advanced features like Hibernation Recon enable the extraction of crucial evidence from memory dumps, notably from hibernation states, significantly aiding in the investigation, particularly in analysing attacker activities (Mehmood, 2023).

3.1.4 CAINE

Computer-Aided Investigative Environment (CAINE) is a Linux Live CD that utilises the robust Ubuntu platform. Unlike other tools like SIFT (SANS Investigative Forensic Toolkit), CAINE is uniquely designed to be directly booted not the target machine, offering an all-encompassing forensic environment from the outset. In this case, the compromised staff PC or other acquired devices. This approach facilitates the seamless integration of new software tools, augmented by a user-friendly graphical interface, making CAINE a versatile and accessible resource for forensic examiners. The emphasis on user-friendliness and versatility streamlines investigative processes, empowering examiners to navigate complex digital environments easily. This comprehensive toolkit and direct deployment capability makes it a valuable asset in the forensic community, enabling to conduct thorough investigations efficiently and effectively (Carranza & DeCusatis, 2021).

3.1.5 ProDiscover

This is a robust computer security tool tailored for law enforcement experts, facilitating comprehensive investigations into laptop disks, focusing on evidence protection and creating high-quality evidentiary reports for criminal proceedings. This disk forensics machine has many capabilities for capturing and analysing disks across a broad spectrum of file systems, including Windows, Linux and Mac. It values employing forensically sound methods throughout the capturing and analysis processes, ensuring integrity and reliability of the findings. Its reports meet rigorous evidentiary standards, providing law enforcement professionals with reliable ad compelling evidence essential for legal proceedings. Focus on thorough analysis, evidence preservation and report generation make it a valuable tool for this investigation, enhancing the capacity to uncover and present compelling evidence in legal settings in case Acme wants to take legal action.

3.2 Analysis

The integration of the tools mentioned enhances the examination and analysis phase, providing a comprehensive toolkit for uncovering evidence related to the keylogger incident. Leveraging the strengths of each tool, investigators can efficiently identify, preserve, and analyse digital evidence, providing Acme with actionable insights. Operational trade-offs such as cost vs, thoroughness and time constraints are carefully considered, however the benefits outweigh the associated costs.

3.2.1 Examination & Analysis Using Forensic Tools

The integration of tools like Autopsy, FTK and Sleuth Kit enhances the examination and analysis, helping uncover evidence related to the Acme Bank's keylogger incident.

Data Triaging: Identify and prioritise data sources based on their potential relevance to the investigation. Utilise FTK's case management features to organise and prioritise evidence sources, focusing on system logs, network traffic logs, and other relevant sources that may contain evidence of the breach.

Forensic Imaging: Utilise FTK Imager to perform forensic imaging of the compromised staff PC, creating a forensic image of the hard drive. This ensures integrity of evidence and preserves the original data for analysis.

Keyword Searching: Use FTK's keyword searching capabilities to search for key terms related to the incident, such as usernames, passwords, and commands associated with the keylogger (Sahran & Yadav, 2022). Perform targeted searches across forensic images and log files to identify relevant evidence.

Timeline Analysis: Utilise Sleuth Kit's timeline analysis capabilities and FTK's timeline feature to construct a chronological timeline of events based on timestamps extracted from system logs, network logs, and other sources. Then analyse the timeline to identify suspicious activities and potential points of compromise.

Artefact Analysis: Use Sleuth Kit and FTK to analyse various artefacts, including file system metadata, registry entries, and memory dumps, to identify traces of the keylogger and any associated malware. Look for indicators of compromise, such as unusual file modifications or unauthorised system changes.

Network Traffic Analysis: Analyse network traffic logs using FTK to identify communication patterns between the compromised PC and external servers. Look for command-and-control (C&C) communication, data exfiltration attempts, or any suspicious network activity.

File Carving: Employ Sleuth Kit or Scalpel's file carving capabilities and FTK's file recovery features to recover hidden or deleted files that may contain evidence of the keylogger installation or usage. File carving techniques reconstruct fragmented files from disk images and identifies potential artefacts.

Pattern Recognition: Using these tools to identify patterns indicative of keylogger activity or malicious behaviour by looking for recurring patterns in system logs, network traffic, and user activity that may signal an ongoing security breach.

Correlation Analysis: Correlate evidence from different sources using Autopsy and FTK to reconstruct the sequence of events leading up to the keylogger discovery. Look for connections between system log entries, network traffic patterns, and user activity to identify potential attack vectors and perpetrators.

Metadata Examination: Examine file metadata using these forensic tools to determine the origin, authorship, and modification history of files associated with the keylogger incident. Look for any discrepancies or anomalies that may indicate tampering or manipulation of evidence.

Documentation and Reporting: Ensure to document all findings, analysis processes and methodologies used during the examination phase utilising Autopsy and FTK's reporting

features. Prepare a detailed report summarising findings of the examination and analysis phase, highlighting key evidence and actionable insights for stakeholders.

3.2.2 Justification

Comprehensive Toolkit: The integration of Autopsy, FTK, Sleuth Kit and Scalpel provides a comprehensive toolkit for the digital forensic investigation, combining strengths of each tool ensures a thorough and practical analysis of Acme's keylogger incident.

Efficiency: FTK's case management and the tools' powerful analysis capabilities enhance the efficiency of the examination and analysis process, enabling investigators to extract, efficiently analyse, and document all evidence related to the incident.

Reliability: These tools are widely recognised in the digital forensic community and known for their reliability and effectiveness in conducting forensic investigations (Banerjee, 2022). Utilising these tools ensures accuracy and credibility of the investigation's findings.

3.2.3 Operational Trade-Offs

Cost vs. Thoroughness: While using Autopsy, FTK, Sleuth Kit and Scalpel may incur initial costs associated with training and software acquisition, their comprehensive capabilities justify the investment by ensuring a thorough and effective investigation.

Time Constraints: Utilising the forensic tools' automation features can help accelerate the examination and analysis process, enabling investigators to meet tight deadlines without compromising quality or integrity of their findings.

3.2.4 Findings

Given the extensive data sources available for analysis and comprehensive capabilities of Autopsy, FTK, Sleuth Kit and Scalpel, the likelihood of finding relevant data related to the keylogger incident is high. By systematically examining and analysing the acquired evidence from the incident at Acme Bank, investigators can uncover key artefacts and provide stakeholders with accurate and actionable insights to mitigate risks and prevent future security incidents.

4. Conclusion

Integrating tools such as Autopsy, FTK, Sleuth Kit and Scalpel into the digital forensic investigation process presents a robust framework for uncovering evidence related to Acme Bank's keylogger incident. By utilising the diverse capabilities of these tools, investigators can efficiently identify, preserve, and analyse digital evidence acquired, providing the bank with actionable insights. The structured approach to this investigation ensures the accuracy and credibility of the findings, supporting informed decision-making. Organisations like Acme Bank can address cyber security incidents effectively by embracing these advanced tools and methodologies.

5. References

- Akbal, E., YAKUT, Ö. F., Dogan, S., TUNCER, T., & Ertam, F. (2021). A Digital Forensics Approach for Lost Secondary Partition Analysis using Master Boot Record Structured Hard Disk Drives. *Sakarya University Journal of Computer and Information Sciences*, *4*(3), 326-346. https://doi.org/10.35377/saucis...1022600
- Al Hosani, H., Yousef, M., Al Shouq, S., & Iqbal, F. (2020, April). State of the art in digital forensics for small scale digital devices. In *2020 11th International Conference on Information and Communication Systems (ICICS)* (pp. 072-078). IEEE. https://doi.org/10.1109/ICICS49469.2020.239531
- Almutairi, A., Satari, B. S., Rivas, C., Stanciu, C. F., Yamani, M., Zohoorsaadat, Z., & Mokhov, S. A. (2020). Evaluation of Autopsy and Volatility for Cybercrime Investigation: A Forensic Lucid Case Study. International Journal of Digital Crime and Forensics (IJDCF), 12(1), 58-89. http://doi.org/10.4018/IJDCF.2020010104
- Banerjee, S., Patil, A., Jadhav, D., & Borkar, G. (2022). Digital forensics as a service: analysis for forensic knowledge. *Cyber Security and Digital Forensics*, 127-162. https://doi.org/10.1002/9781119795667.ch7
- Bhraguram, T. M., Rajakumar, P. S., & Banu, A. A. (2024). Effectiveness of digital forensic investigation through excavation methods of various Linux based tools. *International Journal of Enterprise Network Management*, *15*(1), 70-92. https://doi.org/10.1504/IJENM.2024.137435
- Bhuiyan, R. A., & Czajka, A. (2024). Forensic Iris Image-Based Post-Mortem Interval Estimation. *arXiv preprint arXiv:2404.10172*. https://doi.org/10.48550/arXiv.2404.10172
- Carranza, A., & DeCusatis, C. (2021). Software validation and daubert standard compliance of an open digital forensics model. *Journal of Machine Intelligence and Data Science (JMIDS)*, *2*. https://hal.science/hal-04179975
- Javed, A. R., Ahmed, W., Alazab, M., Jalil, Z., Kifayat, K., & Gadekallu, T. R. (2022). A comprehensive survey on computer forensics: State-of-the-art, tools, techniques, challenges, and future directions. *IEEE Access*, *10*, 11065-11089. https://doi.org/10.1109/ACCESS.2022.3142508

- Khanpara, P., Shah, I., Tanwar, S., Verma, A., & Sharma, R. (2023). Toward the internet of things forensics: A data analytics perspective. *Security and Privacy*, *6*(5), e306. https://doi.org/10.1002/spy2.306
- Mehmood, R. (2023). Volatile Data Acquisition and Analysis by Using Memory Forensics Techniques. *International Journal for Electronic Crime Investigation*, *7*(4), 81-90. https://doi.org/10.54692/ijeci.2023.0704169
- Moric, Z., Redzepagic, J., & Gatti, F. (2021). ENTERPRISE TOOLS FOR DATA FORENSICS. *Annals of DAAAM & Proceedings*. DOI: 10.2507/32nd.daaam.proceedings.014
- Ozcan, S., Astekin, M., Glisson, W. B., & Choo, K. K. R. (2021, October). DIEF: An Autopsy Module for Distributed Identification of E-mail Files from Disk Images. In *2021 IEEE 9th International Conference on Smart City and Informatization (iSCI)* (pp. 53-61). IEEE. https://doi.org/10.1109/iSCI53438.2021.00017
- Saharan, S., & Yadav, B. (2022). Digital and cyber forensics: A contemporary evolution in forensic sciences. In *Crime Scene Management within Forensic Science: Forensic Techniques for Criminal Investigations* (pp. 267-294). Singapore: Springer Singapore. https://doi.org/10.1007/978-981-16-6683-4_11
- Yaacoub, J. P. A., Noura, H. N., Salman, O., & Chehab, A. (2021). Digital forensics vs. Anti-digital forensics: Techniques, limitations and recommendations. *arXiv preprint arXiv:2103.17028*. https://doi.org/10.48550/arXiv.2103.17028